

Individual Assignment 5

Rebecca Martinez

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1. Setup

Packages

```
# general data wrangling and plotting
library(tidyverse)

# file paths
library(here)

# clean column names
library(janitor)

# convert taxon names to snake case
library(snakecase)

# wrap long axis labels
library(scales)

# read Excel files
library(readxl)

# combine plots
library(patchwork)

# custom colors
library(viridis)
```

Data

```
# read in taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# read in aquatic invertebrate data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# read in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# read in water quality data
# na values are included so unusual entries are treated as missing
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

2. Cleaning

NCOS sites

```
# create object with only NCOS quarterly sampling sites
ncos_sites <- sites |>

  # keep sites in the NCOS sector that were sampled quarterly
  filter(sector == "NCOS" &
    sampling_frequency == "Quarterly")
```

Taxon list

```
# clean taxon names so they match the aquatic invertebrate data
taxon_list_clean <- taxon_list |>

  # convert verbatim taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
    case = "snake"))
```

Water quality

```
# create cleaned water quality data frame
water_quality_clean <- water_quality |>

# clean column names
clean_names() |>

# keep only NCOS quarterly sampling sites
semi_join(ncos_sites, by = "site") |>

# create a shared date-site column for joining later
unite("date_site",
      date,
      site,
      remove = TRUE) |>

# group by each date and site combination
group_by(date_site) |>

# calculate median water quality values for each date-site survey
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>

# ungroup data frame
ungroup() |>

# remove known salinity outlier
filter(date_site != "2022-08-12_NVBR")
```

Aquatic invertebrates

```
# create cleaned aquatic invertebrate data frame
aq_ins_clean <- aq_ins |>

# clean column names
clean_names() |>

# keep only NCOS quarterly sampling sites
```

```

semi_join(ncos_sites, by = "site") |>

# rename date column for easier use
rename(date = date_on_vial) |>

# keep date, sample type, site, and major invertebrate taxon columns
select(date,
       sample_type,
       site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>

# keep only filtered beaker samples
filter(sample_type %in% c("FB 250", "FB250")) |>

# convert taxon columns into long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>

# group by date, site, and taxon
group_by(date, site, taxon) |>

# calculate average abundance per liter
summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5,
          .groups = "drop") |>

# create shared date-site column for joining with water quality
unite("date_site",
      date,
      site,
      remove = FALSE) |>

# join water quality data
left_join(water_quality_clean, by = "date_site") |>

```

```

# join taxonomic information
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

# create full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>

# reorder sites by median salinity
mutate(site_full = fct_reorder(site_full,
                              med_sal,
                              .fun = "median",
                              .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

3. Class figures needed for assignment

Site salinity

```

# create object from aq_ins_clean
salinity <- aq_ins_clean |>

# remove rows only if median salinity is missing
filter(!is.na(med_sal)) |>

# create plot
ggplot(mapping = aes(x = site_full,
                     y = med_sal)) +

# show individual salinity measurements for each survey
geom_jitter(height = 0,
            width = 0.1,

```

```

    shape = 21) +

# wrap long site names on x-axis
scale_x_discrete(labels = label_wrap(width = 15)) +

# show median salinity for each site
stat_summary(fun = median,
              geom = "point",
              color = "red",
              shape = 18,
              size = 4) +

# label axes and add title
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +

# use theme different from default
theme_bw() +

# angle x-axis text for readability
theme(axis.text.x = element_text(angle = 45,
                                  hjust = 1))

```

Copepod abundance

```

# create data frame for copepod abundance
copepod_data <- aq_ins_clean |>

# keep only copepods
filter(taxon == "copepod") |>

# log + 1 transform average abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

```

```

# create copepod plot object
copepods <- ggplot(data = copepod_data,

                  # x-axis: site
                  # y-axis: log-transformed average abundance per liter

```

```

        mapping = aes(x = site_full,
                      y = log_ave_lit)) +

# show individual average abundance/L measurements
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +

# show median abundance for each site
stat_summary(fun = median,
            geom = "point",
            color = "blue",
            shape = 18,
            size = 4) +

# wrap long site names on x-axis
scale_x_discrete(labels = label_wrap(width = 15)) +

# label axes and add title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepods") +

# use theme different from default
theme_bw() +

# angle x-axis text for readability
theme(axis.text.x = element_text(angle = 45,
                                  hjust = 1))

```

4. Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan the figure

Goal: create a plot showing differences between sites in ostracod abundance.

- x-axis: `site_full`
- y-axis: `log_ave_lit` (log + 1 transformed average abundance/L)
- geometry to represent average abundance/L: jittered points using `geom_jitter()`

- geometry to represent medians: larger colored diamond points using `stat_summary(fun = median)`

b. Wrangle data

```
# create data frame for ostracod abundance
ostracod_data <- aq_ins_clean |>

# keep only ostracods
filter(taxon == "ostracod") |>

# log + 1 transform average abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows from ostracod data frame
ostracod_data |>
  slice_sample(n = 10)
```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2020-02-21_NVBR	2020-02-21 00:00:00	NVBR	ostr~	4.8	7.34	4.1	6.1
2	2024-01-28_NPB	2024-01-28 00:00:00	NPB	ostr~	3.33	7.43	3.08	6.6
3	2022-11-19_NPB1	2022-11-19 00:00:00	NPB1	ostr~	2	7.26	2.47	3.96
4	2022-01-21_NEC	2022-01-21 00:00:00	NEC	ostr~	0	NA	NA	NA
5	2022-02-21_NPB	2022-02-21 00:00:00	NPB	ostr~	1.33	7.39	34.1	0.34
6	2023-09-12_NVBR	2023-09-12 00:00:00	NVBR	ostr~	0	9.14	NA	20.3
7	2023-05-04_NMC	2023-05-04 00:00:00	NMC	ostr~	1.87	8.32	NA	10.3
8	2023-03-03_NPB2	2023-03-03 00:00:00	NPB2	ostr~	0.133	7.71	1.98	9.59
9	2022-02-23_NVBR	2022-02-23 00:00:00	NVBR	ostr~	0	8.86	45.2	10.2
10	2023-02-28_NPB1	2023-02-28 00:00:00	NPB1	ostr~	0	NA	NA	NA

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code figure

```

# create ostracod plot object
ostracods <- ggplot(data = ostracod_data,

                    # x-axis: site
                    # y-axis: log-transformed average abundance per liter
                    mapping = aes(x = site_full,
                                   y = log_ave_lit)) +

# show individual average abundance/L measurements
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +

# show median abundance for each site with a visually distinct point
stat_summary(fun = median,
            geom = "point",
            color = "purple",
            shape = 18,
            size = 4) +

# wrap long site labels
scale_x_discrete(labels = label_wrap(width = 15)) +

# label axes and add title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +

# use theme different from default
theme_bw() +

# angle x-axis text for readability
theme(axis.text.x = element_text(angle = 45,
                                   hjust = 1))

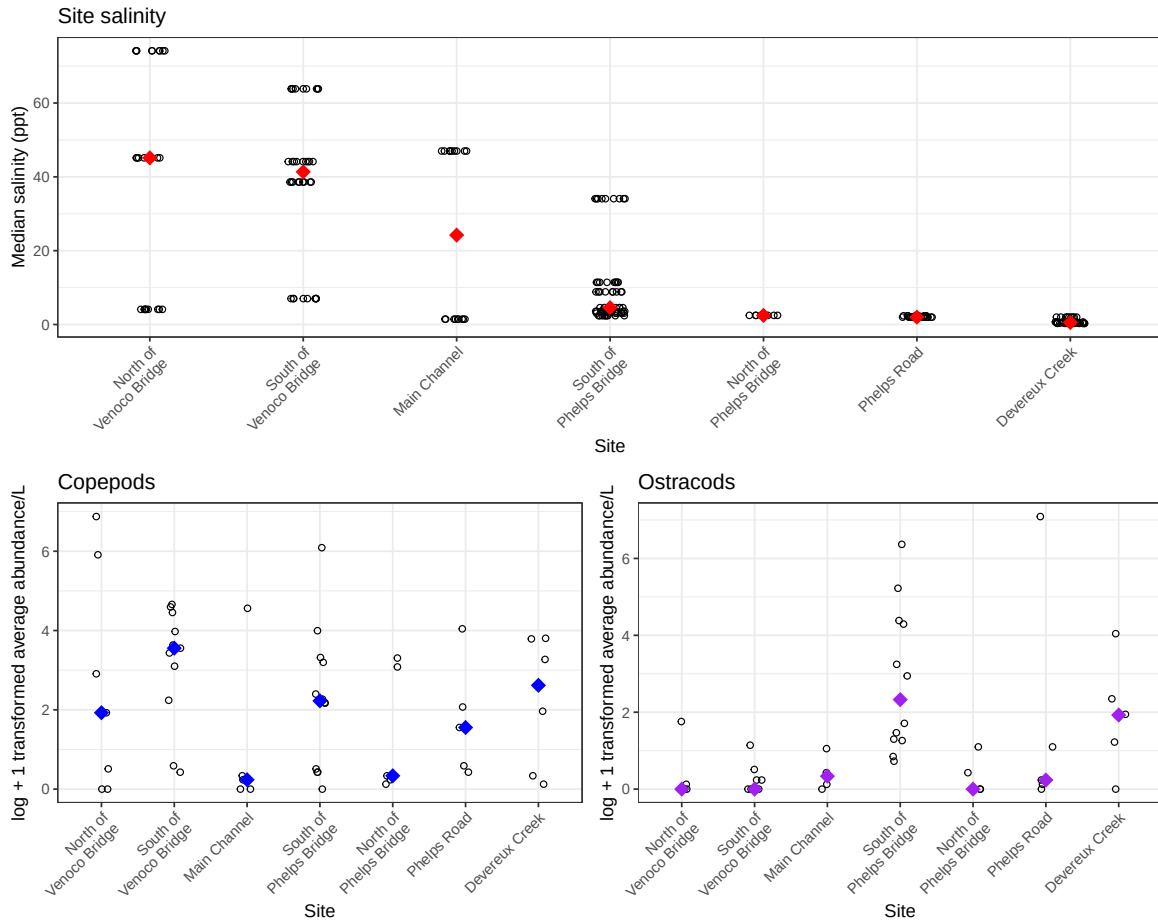
```

d. Create multipanel figure

```

# combine salinity, copepod, and ostracod plots
salinity / (copepods | ostracods)

```



e. Interpretation

Differences between sites in copepod and ostracod abundance: Copepod abundance is relatively high at South of Venoco Bridge, South of Phelps Bridge, and Devereux Creek, shown by the higher blue median points. Ostracod abundance is highest at South of Phelps Bridge and Devereux Creek, where the purple median points are higher than the other sites and several individual observations are also high.

Relationship between ostracod abundance and salinity: Ostracod abundance appears to be higher at lower-salinity sites rather than higher-salinity sites. This is shown by the low red median salinity points at South of Phelps Bridge and Devereux Creek, where ostracod medians are highest, while the Venoco Bridge sites have higher salinity but lower ostracod medians.

2. Visualizing total abundance as a function of salinity

a. Plan the figure

Goal: create a plot showing log-transformed average abundance/L as a function of salinity for each taxon.

- x-axis: `med_sal` (median salinity in ppt)
- y-axis: `log_ave_lit` (log + 1 transformed average abundance/L)
- color: `taxon_title`
- geometry: points using `geom_point()`
- facets: `facet_wrap(~ taxon_title, scales = "free")`

b. Wrangle data

```
# create object for abundance and salinity plot
abundance_salinity <- aq_ins_clean |>

# remove rows only if median salinity is missing
filter(!is.na(med_sal)) |>

# create cleaned columns needed for plotting
mutate(

  # log + 1 transform average abundance per liter
  log_ave_lit = log(ave_lit + 1),

  # make taxon names easier to read
  taxon_title = str_replace_all(taxon, "_", " "),

  # convert taxon names to title case
  taxon_title = str_to_title(taxon_title)
) |>

# order taxa by maximum abundance
mutate(taxon_title = fct_reorder(taxon_title,
                                ave_lit,
                                .fun = max,
                                .desc = TRUE)) |>

# keep only columns needed for figure
```

```

select(date,
       site,
       site_full,
       med_sal,
       taxon,
       taxon_title,
       ave_lit,
       log_ave_lit)

# display 10 random rows from data frame
abundance_salinity |>
  slice_sample(n = 10)

```

A tibble: 10 x 8

	date <dtm>	site <chr>	site_full <fct>	med_sal <dbl>	taxon <chr>	taxon_title <fct>	ave_lit <dbl>
1	2022-05-13 00:00:00	NVBR	North of Venoco ~	74.2	dipt~	Diptera Ce~	0
2	2021-11-23 00:00:00	NPB2	Phelps Road	1.99	nema~	Nematode	0
3	2021-04-15 00:00:00	NPB	South of Phelps ~	3.57	dipt~	Diptera Ch~	1.73
4	2022-02-23 00:00:00	NVBR	North of Venoco ~	45.2	dipt~	Diptera Ch~	0
5	2022-05-05 00:00:00	NEC	South of Venoco ~	63.8	dipt~	Diptera Ch~	0
6	2023-03-03 00:00:00	NPB2	Phelps Road	1.98	anne~	Annelida 0~	0
7	2022-02-25 00:00:00	NEC	South of Venoco ~	44.1	dipt~	Diptera Ce~	0
8	2024-01-28 00:00:00	NPB	South of Phelps ~	3.08	hemi~	Hemiptera ~	0
9	2020-01-21 00:00:00	NDC	Devereux Creek	0.3	ostr~	Ostracod	2.4
10	2023-03-03 00:00:00	NPB2	Phelps Road	1.98	anne~	Annelida P~	0

i 1 more variable: log_ave_lit <dbl>

c. Code your figure

```

# create plot showing abundance as a function of salinity
abundance_salinity_plot <- ggplot(data = abundance_salinity,

                                   # x-axis: median salinity
                                   # y-axis: log-transformed average abundance per liter
                                   # color by taxon
                                   mapping = aes(x = med_sal,
                                                y = log_ave_lit,
                                                color = taxon_title)) +

```

```

# show individual abundance/salinity observations
geom_point() +

# create separate panels for each taxon with free x- and y-axis scales
facet_wrap(~ taxon_title,
            scales = "free") +

# use non-default color palette
scale_color_viridis_d(option = "turbo") +

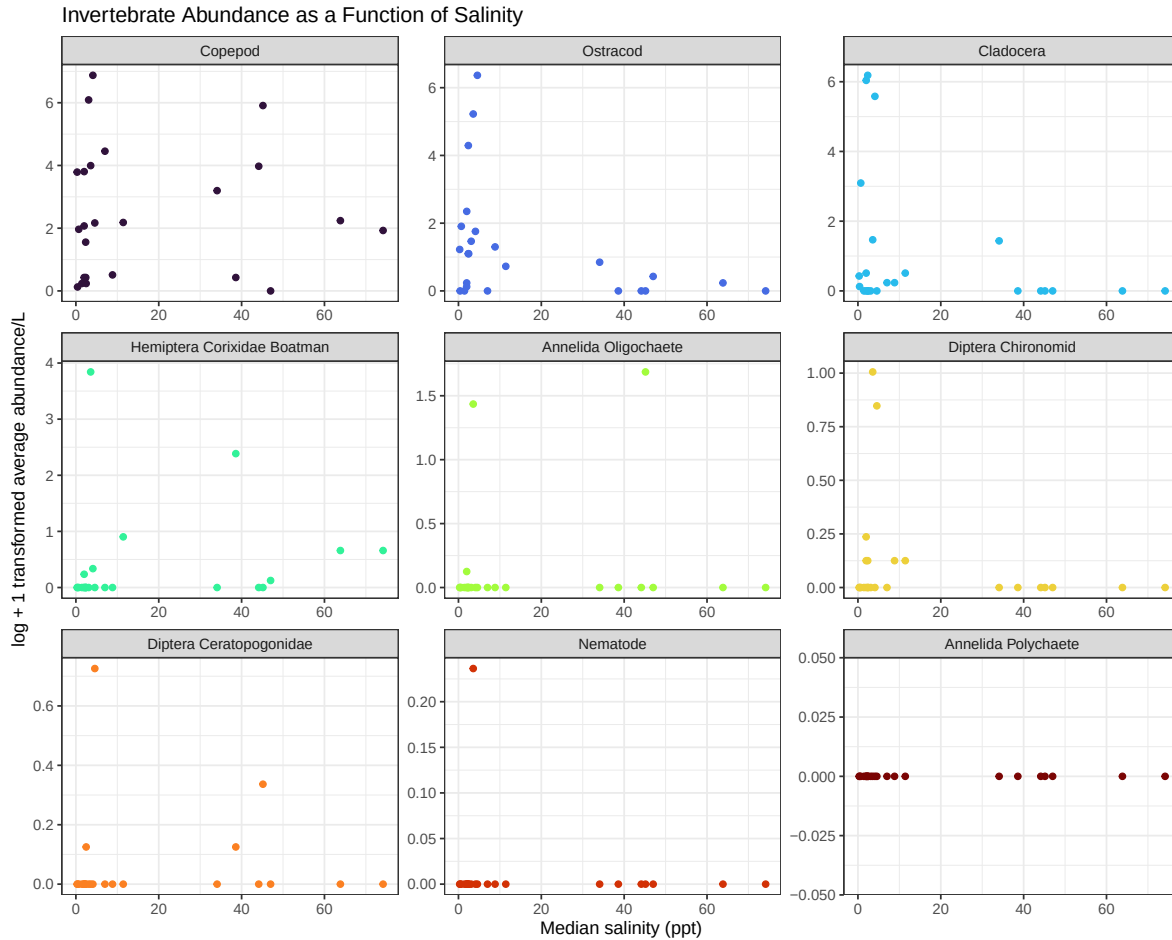
# label x- and y-axes
labs(x = "Median salinity (ppt)",
     y = "log + 1 transformed average abundance/L",
     title = "Invertebrate Abundance as a Function of Salinity") +

# use theme different from default
theme_bw() +

# remove legend because taxa are labeled in facet titles
theme(legend.position = "none")

# display plot
abundance_salinity_plot

```



d. Interpretation

Most taxa have higher abundance at low salinity, shown by points clustered on the left side of the panels where median salinity is around 0–10 ppt. Copepods and ostracods have the highest log-transformed abundance overall, but copepods occur across a wider range of salinity values. Ostracods appear more abundant at lower salinity and lower at higher salinity, suggesting abundance may decrease as salinity increases. Other taxa, like nematodes and annelids, stay low across most salinity values, with points clustered near zero on the y-axis.